

Objectives

To implement and analyze the performance evaluation of a distributed, rule-based heuristic risk scoring algorithm using Apache Spark. The focus is on calculating the baseline accuracy and evaluating true positive classification densities across the distinct student risk tiers using a multiclass confusion matrix under a real-world imbalance.

Tasks Completed

1. Cleaned, prepped, and normalized 1,000,000 student records using distributed Spark RDD operations.
2. Executed a heuristic algorithm applying positive weights to risk factors and negative weights to protective factors.
3. Categorized fixed risk scores into Low, Medium, and High risk tiers using the 33rd and 66th percentiles.
4. Cross-referenced engineered labels against historical ground truths to calculate baseline accuracy and map the confusion matrix.

Tasks to be Completed

N/A

Results and Analysis

1. The algorithm categorized the population into a highly imbalanced distribution: 76.7% Low Risk, 21.8% Medium Risk, and 1.5% High Risk.
2. A baseline model predicting all the students as low risk would yield a 76.7% accuracy. The multiclass confusion matrix resolved this by successfully isolating the true positives within the 1.5% minority tier.
3. Burnout scores (+1.5) and stress levels (+1.4) acted as the strongest mathematical drivers of high-risk classification metrics.

Challenges and Solutions

- Challenge: Standard global accuracy metrics mask algorithm failures in catching dropout due to the 76.7% class imbalance.
- Solution: Drilled down into the density of true positive classifications specifically for the 1.5% minority high-risk tier using a multiclass confusion matrix.

Conclusion

The lab has successfully demonstrated that evaluating big data workflows requires multiclass confusion matrices rather than simple global accuracy. The distributed pipeline successfully isolated the critical 1.5% of high-risk individuals. Establishing these precise validation metrics ensures institutions can confidently deploy data-driven frameworks for targeted, proactive interventions.